

II – Linear Programming Model

Example 1

A farmer wishes to optimize the plantations of rice and corn of his farm, that is, he wants to know which areas to plant of rice and corn in order to achieve the maximum profit of plantations.

The profit per unit of area planted of rice and corn is, respectively, 5 and 2 monetary units (MU).

The areas planted of rice and corn should not be greater than 3 and 4 area units, respectively.

The total consumption of hand labour (measured in man/hours) on both plantations should not be greater than 9. Each area unit planted of rice requires 1 man/hour and each area unit planted of corn needs 2 men/hour.



Decision variables:

- area to plant of rice;
- area to plant of corn.

Objective function:

- maximize the profit obtained from plantations.

Constraints:

- area of the lands;
- hand labour available.



	Rice	Corn	Availability
Area 1	x		3
Area 2		x	4
Hand labour	1	2	9
Profit	5	2	Max

That is, the problem consists in:

Determining

x_1 = area to plant of rice

x_2 = area to plant of corn

in order to

maximize $z = 5x_1 + 2x_2$

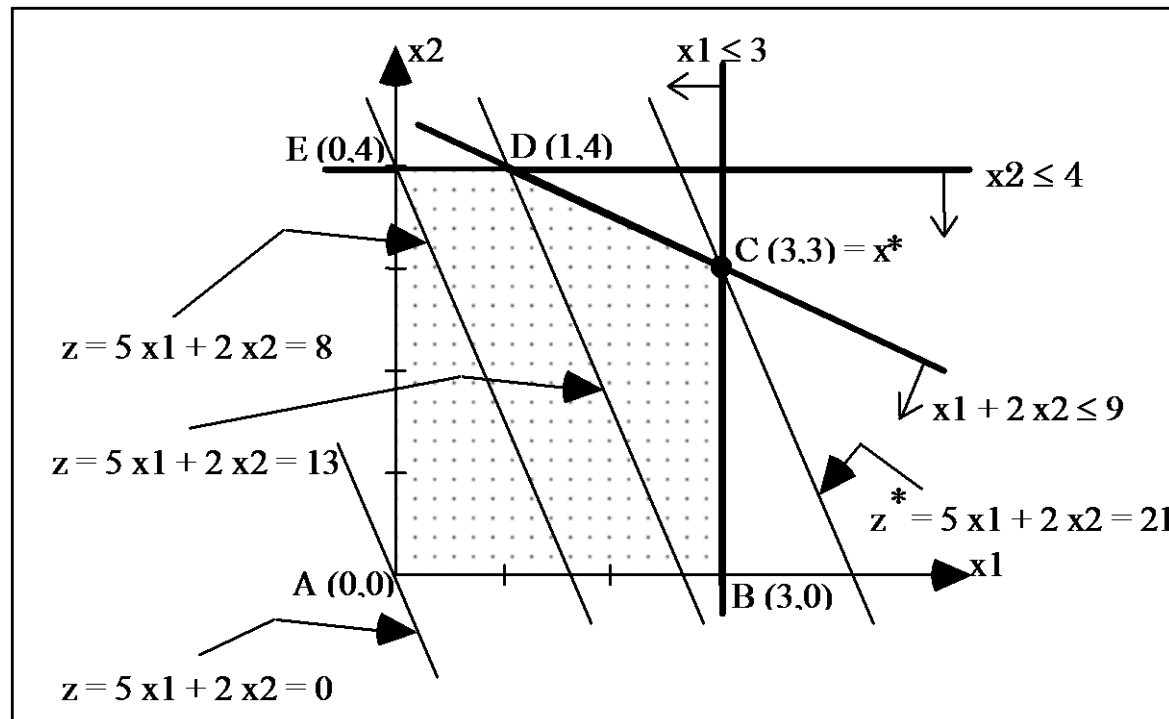
subject to

$$x_1 \leq 3$$

$$x_2 \leq 4$$

$$x_1 + 2x_2 \leq 9$$

$$x_1 \geq 0, x_2 \geq 0$$



Example 2

An office furniture company plans to launch a new model of desks and bookcases.

It is assumed that the market can absorb all the production of bookcases, but it is advised that the monthly production of desks does not exceed 160 units.

Both products are processed in two different units: embossing unit (EU) and assembly and finishing unit (AFU). The monthly availability in each of these units is of 720 machine hours in the EU and of 880 machine hours in AFU. Each desk needs 2 machine hours in EU and 4 machine hours in AFU; in turn, each bookcase needs 4 machine hours in EU and 4 machine hours in AFU.

The profit originated by each desk is of 6 monetary units (MU) and of each shelf is of 3 monetary units (MU).

The company intends to know the monthly production plan of desks and bookcases that maximizes profit.



Decision variables:

- number of desks to produce each month;
- number of bookcases to produce each month.

Objective function:

- maximize the monthly profit.

Constraints:

- availability of EU and AFU;
- market.



Thus, the problem consists in:

Determining

x_1 = number of desks to monthly produce

x_2 = number of bookcases to monthly produce

in order to

maximize $z = 6x_1 + 3x_2$

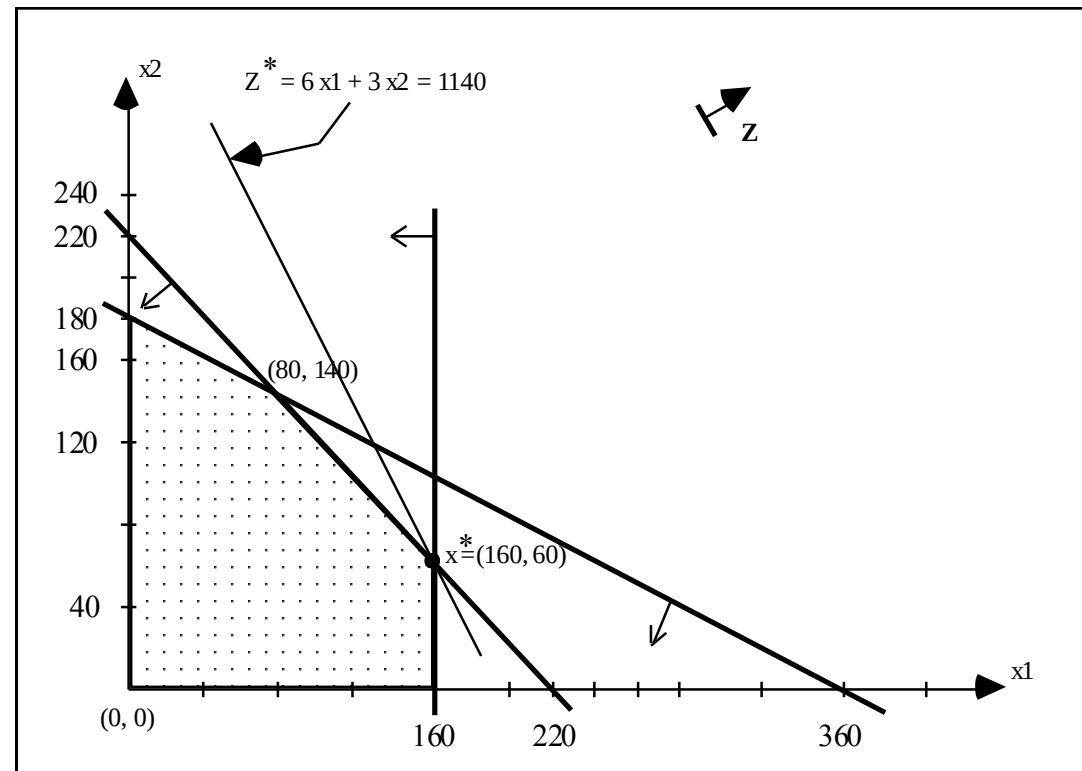
subject to

$$2x_1 + 4x_2 \leq 720$$

$$4x_1 + 4x_2 \leq 880$$

$$x_1 \leq 160$$

$$x_1 \geq 0, x_2 \geq 0$$



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Example 3

A shipping company has a ship with three cargo holds: bow, centre and aft.

Capacity limits of the holds are:



Cargo hold	Tonnage (tons)	Volume (m ³)
Bow	2.000	100.000
Centre	3.200	14.000
Aft	1.800	80.000

Two types of loads (A and B) are offered to the company, each of which it can accept all or part.

Load	Weight (tons)	Volume (m ³ /ton)	Profit (MU/ton)
A	7.000	60	20
B	4.000	25	16

In order to preserve the equilibrium of the ship, the proportion between the weight in each of the holds and their respective tonnages should be kept.

The company wants to know which is the best way to load the ship in order to maximize the profit.



Determine

x_{ij} = number of tons of load i ($i = A, B$) that should be carried in cargo hold j ($j = 1$ ($\rightarrow$ bow),
2 ($\rightarrow$ centre), 3 ($\rightarrow$ aft))

in order to

maximize $z = 20(x_{A1} + x_{A2} + x_{A3}) + 16(x_{B1} + x_{B2} + x_{B3})$

subject to

$$x_{A1} + x_{B1} \leq 2000$$

$$x_{A2} + x_{B2} \leq 3200$$

$$x_{A3} + x_{B3} \leq 1800$$

$$60 x_{A1} + 25 x_{B1} \leq 100000$$

$$60 x_{A2} + 25 x_{B2} \leq 14000$$

$$60 x_{A3} + 25 x_{B3} \leq 80000$$

$$\frac{x_{A1} + x_{B1}}{2000} = \frac{x_{A2} + x_{B2}}{3200} = \frac{x_{A3} + x_{B3}}{1800}$$

$$x_{A1} + x_{A2} + x_{A3} \leq 7000$$

$$x_{B1} + x_{B2} + x_{B3} \leq 4000$$

$x_{ij} \geq 0$ for all i and all j



Linear Programming Model

Generically a linear programming problem can be formulated as follows:

a) Cartesian Form

Determine

$$x_1, x_2, \dots, x_j, \dots, x_n \geq 0$$

in order to

$$\text{maximize } z = c_1 x_1 + c_2 x_2 + \dots + c_j x_j + \dots + c_n x_n$$

subject to

$$a_{11} x_1 + a_{12} x_2 + \dots + a_{1j} x_j + \dots + a_{1n} x_n \leq b_1$$

$$a_{21} x_1 + a_{22} x_2 + \dots + a_{2j} x_j + \dots + a_{2n} x_n \leq b_2$$

$$\cdot \quad \quad \quad \cdot \quad \quad \quad \cdot$$

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$$a_{i1} x_1 + a_{i2} x_2 + \dots + a_{ij} x_j + \dots + a_{in} x_n \leq b_i$$

$$\cdot \quad \quad \quad \cdot \quad \quad \quad \cdot$$

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$$a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mj} x_j + \dots + a_{mn} x_n \leq b_m$$

b) Matricial Form

Determine

$$\mathbf{x} \geq \mathbf{0}$$

in order to

$$\text{maximize } z = \mathbf{c}'\mathbf{x}$$

subject to

$$\mathbf{Ax} \leq \mathbf{b}$$

Where:

$\mathbf{x}$ – vector $n \times 1$

$\mathbf{0}$ – vector $n \times 1$

z – scalar

$\mathbf{c}$ – vector $n \times 1$

$\mathbf{A}$ – matrix $m \times n$

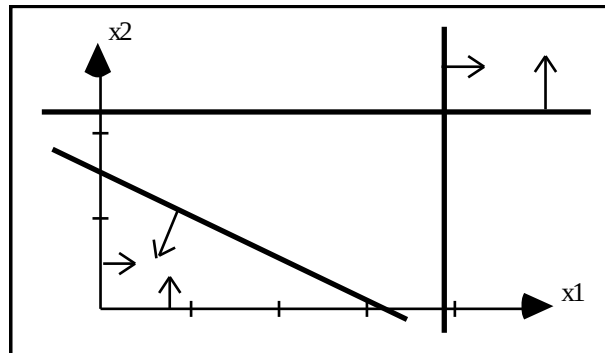
$\mathbf{b}$ – vector $m \times 1$

Note that:

- the type of optimization may be a *minimization*;
- the constraints may be of type " $\geq$ " or " $=$ ";
- some of the decision variables may not satisfy the condition of being " ≥ 0 ".

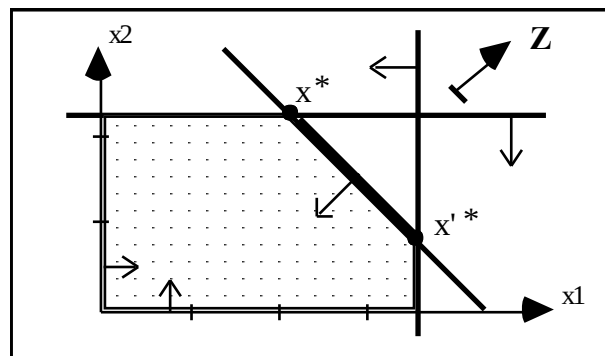
Particular cases

- **Nonexistent solution**
(there is no feasible region)



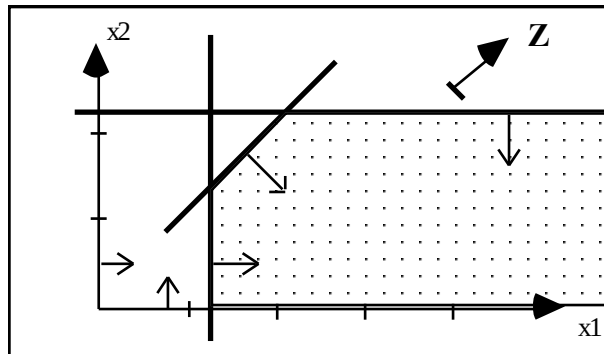
- **Alternative optimal solutions**

(the optimal value of the objective function can be achieved by multiple combinations of resources)



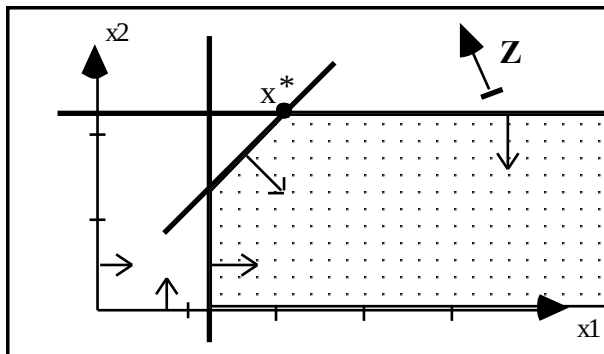
➤ **Unbounded solution**

(non-finite optimal value of the objective function)



➤ **Optimal solution**

(despite the feasible region is unbounded)



➤ **Finite optimal solution**

(with variables assuming arbitrarily large values)

